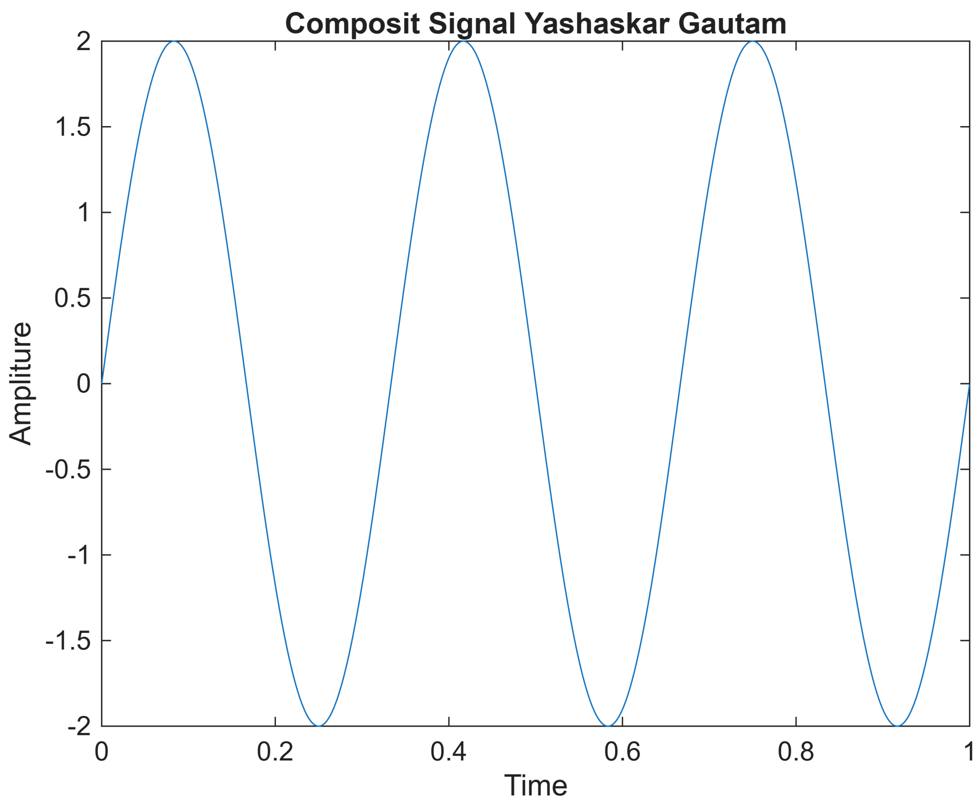
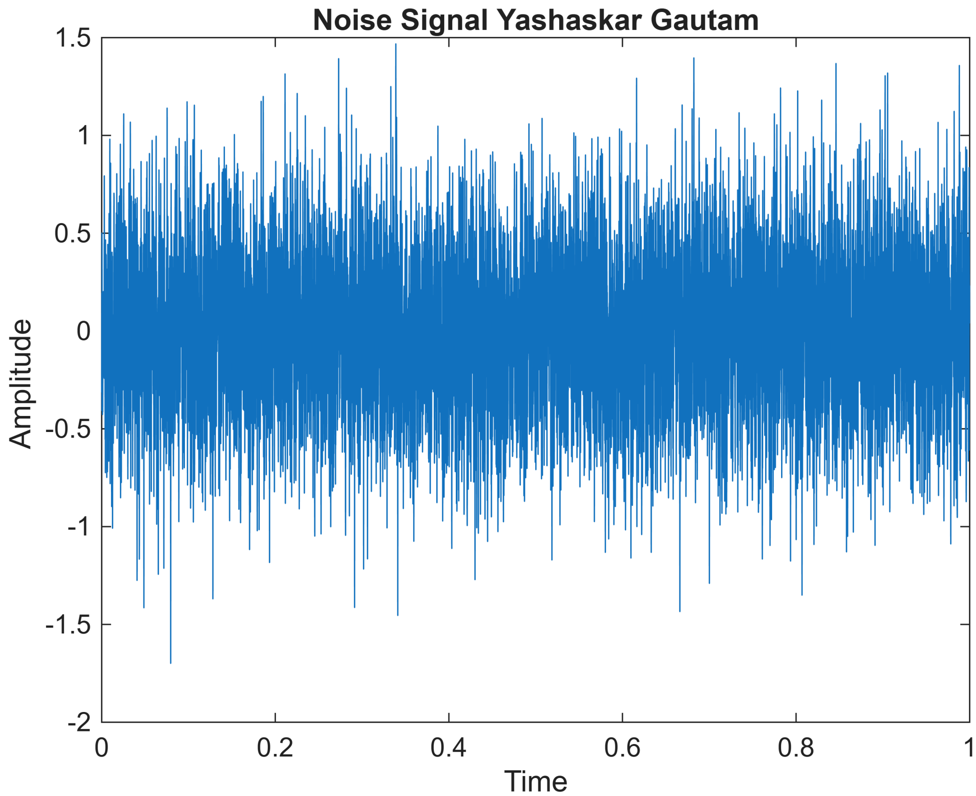
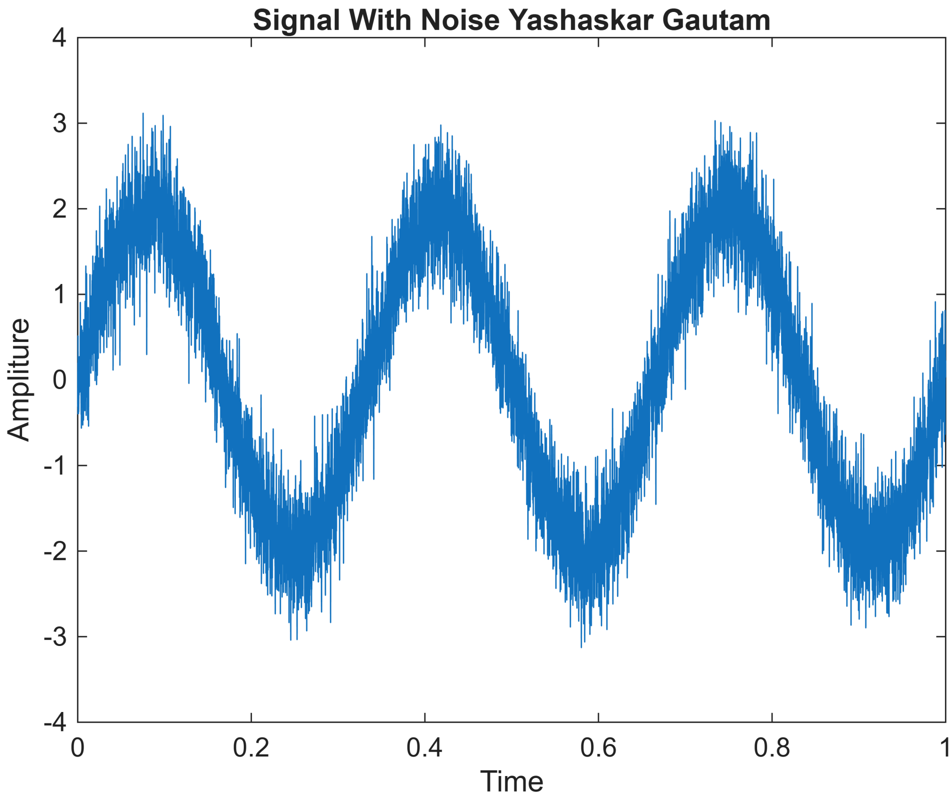
```
% Shannon Capicity Calculation for Noise Channel


clear all;
clc;

fs = 8000; % Sampling frequency
t = 0:1/fs : 1-1/fs; % Time duration
%cx = 1.1*sin(2*pi*100*t) + 1.3*cos(2*pi*300*t) + 1.5*sin(2*pi*2000*t);
cx = 2*sin(2*pi*3*t);

plot(t,cx);
xlabel('Time')
ylabel('Ampliture')

title('Composit Signal Yashaskar Gautam');
```



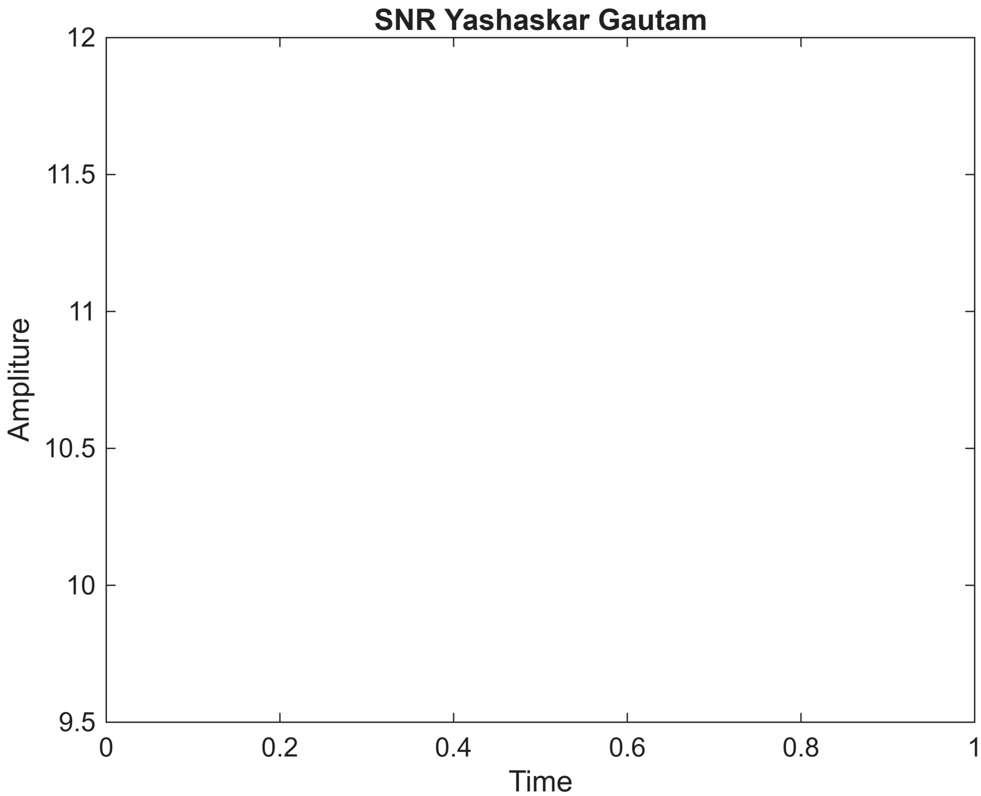

```
ns =  0.4 *randn(size(cx));

plot(t,ns);
xlabel('Time')
ylabel('Amplitude ')

title('Noise Signal Yashaskar Gautam');
```




```
% signal with noise

S_W_N = cx + ns;

plot(t,S_W_N);
xlabel('Time')
ylabel('Ampliture')

title('Signal With Noise Yashaskar Gautam');
```




```
S_N_R = snr(S_W_N,fs);

plot(t,S_N_R);
xlabel('Time')
ylabel('Ampliture')

title('SNR Yashaskar Gautam');
```




```
bandwidth = obw(cx,fs); % Bandwidth of the signal
L=2; % Level of the signal
C = bandwidth*log2(1+S_N_R)
```

```
C =
3.5349
```